



Fig. 2: Predicting market trends

like mobile services, cable TV and broadcasting.

The various government initiatives like Make in India and the development of smart cities are likely to boost power and communications sectors in India and eventually contribute to the T&M industry's growth.

The demand for electricity across India has been growing at a rapid rate and is expected to grow further in the years to come. With the growing demand, power generation, transmission and distribution companies will have to invest in reliable, accurate and efficient equipment to protect their assets from failure. Therefore the growth prospects of T&M equipment related to the power sector are quite promising. India is an emerging solar hub of the world, with nearly 100GW of solar power to be added in the next five years. Solar power, again, is a very important market for T&M products.

The trend of automotive companies switching to automated testing is also driving up demand for T&M equipment. Moreover, the increasing electronic content in vehicles for infotainment, connectivity and safety purposes, along with the growing acceptance of green mobility (through e-vehicles), will fuel the growth of the T&M sector.

Quality and safety norms compliance

- Over-voltage safety standards like CAT IV, III, II and I, which ensure the instrument is safe under different working environments
- Ingress protection (IP) standards for protection against dust and water
- IEC 61010-1

Market trends

Survey participants listed the following emerging trends shaping the T&M market:

Support for predictive maintenance. Indian manufacturers' increasing awareness about predictive maintenance is driving the growth of the T&M industry.

Equipment with unique features. Customer demand to address complex process issues is forcing T&M companies to innovate continuously. This has resulted in the development of high-end devices with unique features.

Multi-functional and multi-domain testing. Maintenance engineers have realised the benefits of using one kit with multi-functional testing capabilities instead of having many different testing kits for individual applications. Such tools have multi-domain applications as well.

Smart testing for 'smart devices'. For both testing managers and engineers, the challenge is to ensure quality at increasingly lower costs. To test their smart devices, organisations are transitioning from the status quo of rack-and-stack box instruments and closed turnkey automated test equipment to smarter test systems with connectivity and problem-solving capabilities.

Customised equipment. The current focus on customised test equipment is driving the overall market, as most of the low-end T&M equipment are now available with control options.

Quality ensures long-term success. According to a majority of the survey respondents, customers

are willing to spend extra for good quality. However, the definition of quality can be subjective and often determined by the customer's unique perspective. Nevertheless, survey respondents believe that objective standards of quality lead to quantifiable benchmarks, as every customer segment wants its systems to function properly and reliably on a consistent basis.

A few experts mentioned that in the telecom sector, huge investments have been made in spectrum procurement but decision makers are not willing to invest in T&M equipment. Instead, mobile network operators have adopted the rental, managed services or capex model. This reduces the need to procure T&M equipment but can also lead to poor service or product quality.

Impact of emerging 'tech' trends

Connected life-style, machine-to-machine (M2M) communication and the Internet of Things (IoT) will drive the growth of the T&M industry.

The proliferation of smart devices has begun to influence the design and use of T&M equipment. Each 'smart' product has to go through a specific set of processes to ensure its usability and functionality for the intended purpose. Behind each of these steps are the design, manufacturing and testing procedures. T&M technologies are going to evolve accordingly.

As the time-to-market for new devices keeps shortening, maintaining high quality will continue to drive additional complexity in these devices—and all this will be invisible to users. Test techniques and test solutions will continue to evolve, enabling smart device developers and manufacturers to reduce time-to-market and ramp-up manufacturing of quality products quickly.

In the telecom market, the need for high-speed connectivity will lead to increase of the network capacity. Operators are using newer technolo-